

CSS Layout - Flexbox – Lab

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Topics Covered

[Flexbox](#) is a one-dimensional layout method for arranging items in rows or columns. Items *flex* (expand) to fill additional space or shrink to fit into smaller spaces. This article explains all the fundamentals.

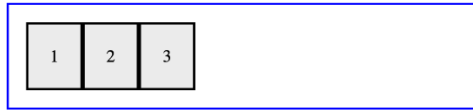
Resources

- **Flexbox MDN**
 - https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Flexbox
- **A Complete Guide to Flexbox**
 - <https://css-tricks.com/snippets/css/a-guide-to-flexbox/>
- **An Interactive Guide to Flexbox**
 - <https://www.joshwcomeau.com/css/interactive-guide-to-flexbox/>
- **Flexbox cheat sheet**
 - <https://codepen.io/kdankov/pen/QWoVyZl>

Tasks

1. Property – display: flex;

display: flex



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **display-flex.html**
- Create a **CSS** file named **display-flex.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create a **<div>** element with a class – "**flex-container**"
- Create 3 **<div>** elements inside the .flex-container element with a class of "**flex-item**"
- Add text to the .flex-item elements as shown on the screenshot

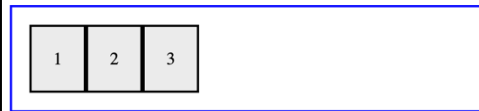
3. Styling

- Add the following styles the .flex-container element:
 1. Width – 400px
 2. Padding – 1em
 3. Border – 2px solid blue
 4. Display – flex
- Add the following styles to the .flex-item elements inside the .flex-container element:
 1. Background - #eee
 2. Padding – 20px
 3. Border – 2px solid #000

2. Property – flex-direction

flex-direction

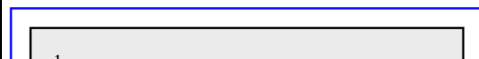
flex-direction: row; (default)



flex-direction: row-reverse;



flex-direction: column;



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **flex-direction.html**
- Create a **CSS** file named **flex-direction.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 4 sets of **<h2>** element with a **<div>** element with a class – "**flex-container**" as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of "**flex-item**"
 2. Add text to the .flex-item elements as shown on the screenshot

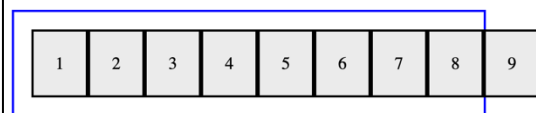
3. Styling

- Copy the styles from Task 1
- Create a new class "**flex-direction-row**" with the following styles
 1. Flex-direction – row
- Create a new class "**flex-direction-row-reverse**" with the following styles
 1. Flex-direction – row-reverse
- Create a new class "**flex-direction-column**" with the following styles
 1. Flex-direction – column
- Create a new class "**flex-direction-column-reverse**" with the following styles
 1. Flex-direction – column-reverse
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

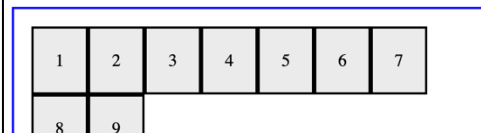
3. Property – flex-wrap

flex-wrap

flex-wrap: nowrap; (default)



flex-wrap: wrap;



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **flex-wrap.html**
- Create a **CSS** file named **flex-wrap.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 3 sets of **<h2>** element with a **<div>** element with a class – "**flex-container**" as shown in the screenshot
 1. Create **<div>** elements inside the **.flex-container** element with a class of "**flex-item**"
 2. Add text to the **.flex-item** elements as shown on the screenshot

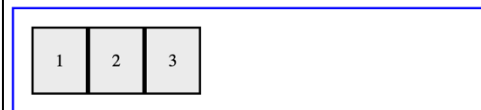
3. Styling

- Copy the styles from Task 1
- Create a new class "**flex-wrap-nowrap**" with the following styles
 1. Flex-wrap – nowrap
- Create a new class "**flex-wrap-wrap**" with the following styles
 1. Flex-wrap – wrap
- Create a new class "**flex-wrap-wrap-reverse**" with the following styles
 1. Flex-wrap – wrap-reverse
- Apply the classes to the appropriate **.flex-container** elements to achieve the results shown in the screenshot

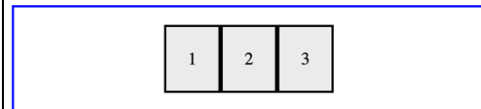
4. Property – justify-content

justify-content

justify-content: flex-start; (default)



justify-content: center;



justify-content: flex-end;



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **justify-content.html**
- Create a **CSS** file named **justify-content.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 6 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the **.flex-container** element with a class of **"flex-item"**
 2. Add text to the **.flex-item** elements as shown on the screenshot

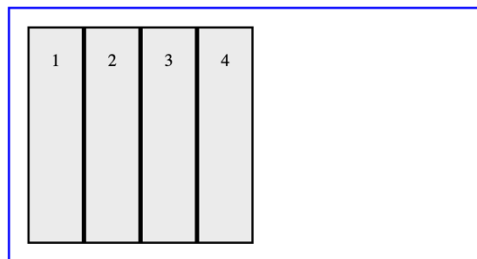
3. Styling

- Copy the styles from Task 1
- Create a new class **"justify-content-flex-start"** with the following styles
 1. justify-content – flex-start
- Create a new class **"justify-content-center"** with the following styles
 1. justify-content – center
- Create a new class **"justify-content-flex-end"** with the following styles
 1. justify-content – flex-end
- Create a new class **"justify-content-space-between"** with the following styles
 1. justify-content – space-between
- Create a new class **"justify-content-space-around"** with the following styles
 1. justify-content – space-around
- Create a new class **"justify-content-space-evenly"** with the following styles
 1. justify-content – space-evenly
- Apply the classes to the appropriate **.flex-container** elements to achieve the results shown in the screenshot

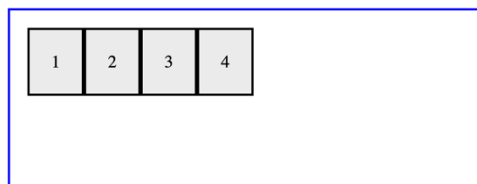
5. Property – align-items

align-items

align-items: stretch; (default)



align-items: flex-start;



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **align-items.html**
- Create a **CSS** file named **align-items.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 5 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the **.flex-container** element with a class of **"flex-item"**
 2. Add text to the **.flex-item** elements as shown on the screenshot

3. Styling

- Copy the styles from Task 1
- Add the following styles to the **.flex-container** class:
 1. height - 200px
- Create a new class **"align-items-stretch"** with the following styles
 1. align-items – stretch
- Create a new class **"align-items-flex-start"** with the following styles
 1. align-items – flex-start
- Create a new class **"align-items-center"** with the following styles
 1. align-items – center
- Create a new class **"align-items-flex-end"** with the following styles
 1. align-items – flex-end
- Create a new class **"align-items-baseline"** with the following styles
 1. align-items – baseline

- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

6. Property: Align Content

Use the skeleton is in resources to achieve the following view:

align-content

align-content: flex-start; (default)

1	2	3	4	5	6	7
8	9	10	11	12		

align-content: center;

For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **align-content.html**
- Create a **CSS** file named **align-content.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 5 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of **"flex-item"**
 2. Add text to the .flex-item elements as shown on the screenshot

3. Styling

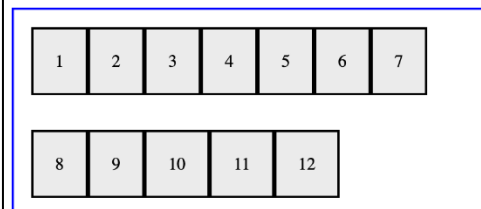
- Copy the styles from Task 1
- Add the following styles to the .flex-container class:
 1. height - 200px
 2. align-items - flex-start
 3. flex-wrap - wrap
- Create a new class **"align-content-flex-start"** with the following styles
 1. align-content – flex-start
- Create a new class **"align-content-center"** with the following styles
 1. align-content – center
- Create a new class **"align-content-flex-end"** with the following styles
 1. align-content – flex-end
- Create a new class **"align-content-space-between"** with the following styles
 1. align-content – space-between

- Create a new class **"align-content-space-around"** with the following styles
 1. align-content – space-around
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

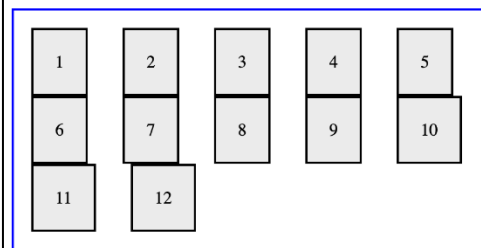
7. Property: Gap

gap

gap row (gap: 2em 0;)



gap column (gap: 0 2em;)



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

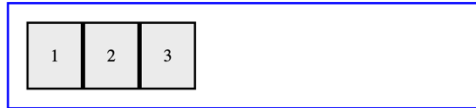
1. **File Setup:**
 - Create an **HTML** document named **gap.html**
 - Create a **CSS** file named **gap.css**
2. **Element creation**
 - Add title of the page in an **<h1>** element
 - Create 3 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of **"flex-item"**
 2. Add text to the .flex-item elements as shown on the screenshot
3. **Styling**
 - Copy the styles from Task 1
 - Add the following styles to the .flex-container class:
 1. align-items – flex-start
 2. flex-wrap - wrap
 - Create a new class **"gap-row"** with the following styles
 1. gap - 2em 0;
 - Create a new class **"gap-column"** with the following styles
 1. gap - 0 2em;
 - Create a new class **"gap"** with the following styles

1. gap - 2em;
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

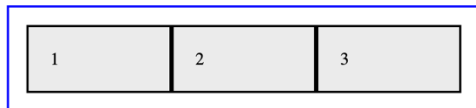
8. Property: Flex Grow

flex-grow

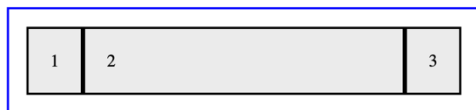
flex-grow: 0; (default)



flex-grow: 1;



flex-grow, grow 2nd child only



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. File Setup:

- Create an **HTML** document named **flex-grow.html**
- Create a **CSS** file named **flex-grow.css**

2. Element creation

- Add title of the page in an **<h1>** element
- Create 4 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of **"flex-item"**
 2. Add text to the .flex-item elements as shown on the screenshot

3. Styling

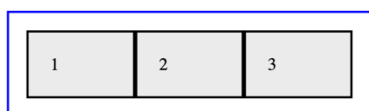
- Copy the styles from Task 1
- Add the following styles to the .flex-container class:
 1. align-items – flex-start
 2. flex-wrap - wrap
- Create a new class **"flex-grow-one"** and use it to style all .flex-item elements inside it with the following styles
 1. flex-grow - 1;
- Create a new class **"flex-grow-2nd-only"** and use it to style the 2nd .flex-item element inside it with the following styles
 1. flex-grow - 1;

- Create a new class **"flex-grow-proportional"** and use it to style each .flex-item element inside it with the following styles
 1. Element 1
 1. flex-grow - 1;
 2. Element 2
 1. flex-grow - 2;
 3. Element 3
 1. flex-grow - 3;
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

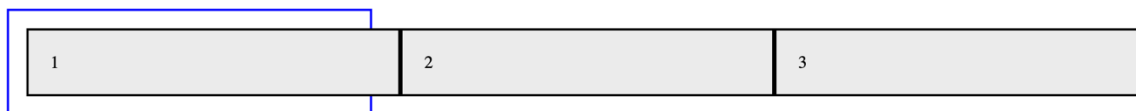
9. Property: Flex Shrink

flex-shrink

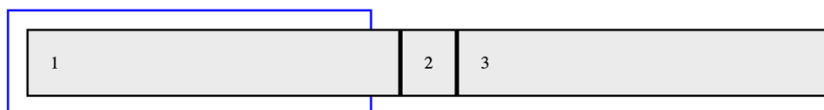
flex-shrink: 1; (default)



flex-shrink: 0;



flex-shrink, shrink 2nd child only



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

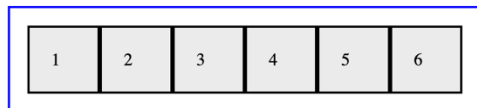
1. **File Setup:**
 - Create an **HTML** document named **flex-shrink.html**
 - Create a **CSS** file named **flex-shrink.css**
2. **Element creation**
 - Add title of the page in an **<h1>** element
 - Create 3 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of **"flex-item"**
 2. Add text to the .flex-item elements as shown on the screenshot
3. **Styling**
 - Copy the styles from Task 1
 - Add the following styles to the .flex-item class:
 1. flex-basis – 300px

- Add the following styles to the `.flex-container` class:
 1. `align-items – flex-start`
 2. `width – 300px`
- Create a new class **"flex-shrink-zero"** and use it to style all `.flex-item` elements inside it with the following styles:
 1. `flex-shrink - 0;`
- Create a new class **"flex-shrink-2nd-only"** and use it to style all `.flex-item` element inside it with the following styles:
 1. `flex-shrink - 0;`
- Use the class **"flex-shrink-2nd-only"** to style the 2nd `.flex-item` element inside it with the following styles:
 1. `flex-shrink - 1;`
- Apply the classes to the appropriate `.flex-container` elements to achieve the results shown in the screenshot

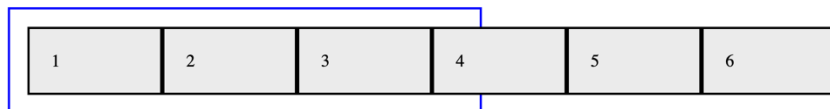
10.Property: Flex Basis

flex-basis

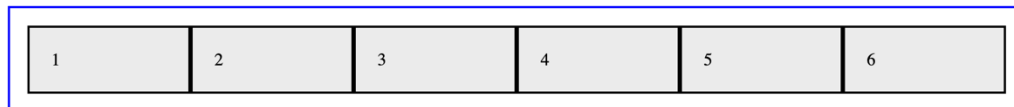
flex-basis: 5em;



flex-basis: 5em; flex-shrink: 0; (max-width: 5em)



flex-basis: 5em; flex-grow: 1; (min-width: 5em)



For full screenshot check the `screenshot.png` file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

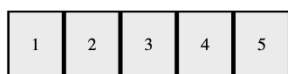
1. **File Setup:**
 - Create an **HTML** document named **flex-basis.html**
 - Create a **CSS** file named **flex-basis.css**
2. **Element creation**
 - Add title of the page in an **<h1>** element
 - Create 3 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the `.flex-container` element with a class of **"flex-item"**
 2. Add text to the `.flex-item` elements as shown on the screenshot
3. **Styling**

- Copy the styles from Task 1
- Create a new class **"flex-basis-5em"** and use it to style all .flex-item elements inside it with the following styles:
 1. flex-basis - 5em;
- Create a new class **"flex-shrink-zero"** and use it to style all .flex-item element inside it with the following styles:
 1. flex-shrink - 0;
- Create a new class **"flex-grow-one"** with the following styles:
 1. Width - 900px;
- Use the class **"flex-grow-one"** it to style all .flex-item element inside it with the following styles:
 1. flex-grow - 1;
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot

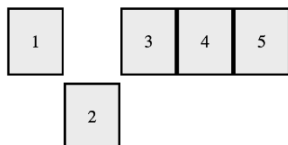
11.Property: Align Self

align-self

align-self: flex-start;



align-self: center;



For full screenshot check the screenshot.png file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. **File Setup:**
 - Create an **HTML** document named **align-self.html**
 - Create a **CSS** file named **align-self.css**
2. **Element creation**
 - Add title of the page in an **<h1>** element
 - Create 4 sets of **<h2>** element with a **<div>** element with a class – **"flex-container"** as shown in the screenshot
 1. Create **<div>** elements inside the .flex-container element with a class of **"flex-item"**
 2. Add text to the .flex-item elements as shown on the screenshot
3. **Styling**

- Copy the styles from Task 1
- Add the following styles to the `.flex-container` class:
 1. `align-items – flex-start`
- Create a new class **"align-self-flex-start"** and use it to style the 2nd `.flex-item` element inside it with the following styles:
 1. `align-self - flex-start;`
- Create a new class **"align-self-center"** and use it to style the 2nd `.flex-item` element inside it with the following styles:
 1. `align-self - center;`
- Create a new class **"align-self-flex-end"** and use it to style the 2nd `.flex-item` element inside it with the following styles:
 1. `align-self - flex-end;`
- Create a new class **"align-self-baseline"** and use it to style the 2nd `.flex-item` element inside it with the following styles:
 1. `align-self - baseline;`
- Apply the classes to the appropriate `.flex-container` elements to achieve the results shown in the screenshot

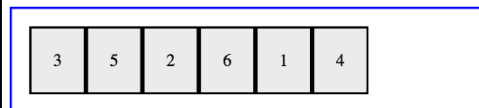
12.Property: Order

order

Default Order



Custom Order



For full screenshot check the `screenshot.png` file in the folder for the task

Objective:

Create a webpage that looks like the screenshot above.

Requirements:

1. **File Setup:**
 - Create an HTML document named **order.html**
 - Create a CSS file named **order.css**
2. **Element creation**
 - Add title of the page in an `<h1>` element
 - Create 2 sets of `<h2>` element with a `<div>` element with a class – **"flex-container"** as shown in the screenshot
 1. Create `<div>` elements inside the `.flex-container` element with a class of **"flex-item"**
 2. Add text to the `.flex-item` elements as shown on the screenshot

3. Styling

- Copy the styles from Task 1
- Add the following styles to the .flex-container class:
 1. align-items – flex-start
- Create a new class “custom-order” and use it to style each individual .flex-item element inside it with the following styles:
 1. Item 1
 1. Order – 5
 2. Item 2
 1. Order – 3
 3. Item 3
 1. Order – 1
 4. Item 4
 1. Order – 6
 5. Item 5
 1. Order – 2
 6. Item 6
 1. Order – 4
- Apply the classes to the appropriate .flex-container elements to achieve the results shown in the screenshot